AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M269
Algorithms, Data Structures and Computability

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 18 /Nov./2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **60 marks**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer Booklet.
- Be sure you write your **name and ID** on the External Answer Booklet.
- **Calculators** are not allowed.

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PART 1: ALL QUESTIONS ARE REQUIRED [10 Marks]

Question 1: (10 marks)

For the following statements please answer "True" or "False":

1) Using bubble sort on an array A[n] involves (log n) comparisons.
2) retrieving the first element in a Stack of size n is of O(1)
3) O(log n) is less in complexity than O(n)
4) In a queue, If we implement enqueue(8), enqueue(2), dequeue(), enqueue(2), enqueue(7), dequeue(), dequeue(), and enqueue(7) the remaining value in the queue will be 7.
5) If the "insertion-sort" is implemented on the array A[44, 25, 12, 61, 31], after the first two cycles the array will be A[12, 25, 44, 61, 31]
6) The best case complexity of Bubble sort algorithm is $O(n^2)$
7) The average complexity of merge sort algorithm is O(n log n)
8) The tree data-structure can be used to represent hierarchical relationship between elements.
9) When determining the efficiency of an algorithm, the space factor is measured by counting the maximum memory needed by the algorithm.
10) Linked lists are the best data-structure in case the size and structure of the data are constantly changing.

PART 2: ALL QUESTIONS ARE REQUIRED [25 Marks]

Question 2: (13 marks)

a) Explain the performance of the Bubble-sort and Heap-sort in case they are implemented on the following inputs. [Your answer should give the complexity of both algorithms and decide which of them will be better than the other for each case]:

(4 marks: 2 marks each)

1) An array of n items all of the same value.
2) An array of n items that are sorted in reverse order

b) What is the complexity of the following tasks? (6 marks: 1 mark each)

1) Finding the largest element in a queue of size n+3
2) Inserting an item in a queue data-structure, which has already n element.
3) Computing the mathematical average of all items in a queue of size 2n
4) Sorting a sorted array using quick sort algorithm of size n
5) Build maximum heap () on any given unsorted array of size n
6) Dequeue the last task inserted into an array of size n

c) What is the best data structure that suits each of the following task (3 marks: 1 mark each)

1) Revisit the last webpage visited in a browser
2) Serving a series of tasks in the computer processor
3) Presenting the files and folders on a computer drive

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Question 3:(12 marks)

Consider the list = [45,28,50,18,41,33] and answer the below questions

This list has been sorted using only **three sorting** algorithm out of the following ones: Insertion-Sort, Merge-Sort, Bubble-Sort, Quick-Sort and Selection-Sort. The output of the above list after the **2nd cycle** of implementing the three randomly selected sorting algorithms (each one apart) is as follows:

i) 28,18,41,33,45,50
ii) 18,28,50,45,41,33
iii) 28,45,50,18,41,33

a) Determine which sorting algorithm has been used in each of the above cases. (3 marks)
b) Conduct the 3rd cycle using the same sorting algorithm in each case and show the output. (6 marks total: 2 marks each)
c) Show the list after implementing the Max-Heapify algorithm. (3 marks)

PART 3: ALL QUESTIONS ARE REQUIRED [25 marks]

Question 4: (15 marks)

a) Find the complexity of the following algorithms, where A, B are arrays, n is the input size [Note: your answer should include steps that derive to the final answer] (total 6 marks: 3 marks each)

1)	2)
for i = 1 to n Do A[i] = 1 / i B[i] = i^2 endfor InsertionSort(A, n) BubbleSort(B, n)	i = 1 while i < n do i = 2 × i endwhile j =1 while j < 10 do j = 2 × j endwhile

b) Use Master Theorem to find the O notation for the following recurrence relations. [Note: your answer should include steps that derive to the final answer] (total 6 marks: 3 marks each)

1) $T(n) = 16T(n/4) + n$
2) $T(n) = T(n/2) + n$

c) Fill the table below with the Big-O notation and the order (which is a number from 1 to 5, where 1 represents the asymptotically smallest and 5 represents the asymptotically largest): (Total 5 marks: 0.5 mark per each cell)

Expression	Big-O	Order
$4\ n^2 \times 4\ \log n$		
$\log n / 4 + 4\ n^3$		
$10\ n + 9\ n \log 8$		
$3\ n^2 + 3n^3$		
$50\ n^3 \log 4n$		

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Question 5: (10 marks)

Given an unsorted integer array whose elements lie in the range 0 to n-1 where n is the array size, calculate the frequency of all array elements in linear time and using constant space.

For example
Input: { 2, 3, 3, 2, 1 }
Output:
Element 1 appears once.
Element 2 appears twice.
Element 3 appears twice.

a) Design an efficient algorithm to solve this problem as illustrated in the two given examples (6 marks)
b) Analyze the complexity of your algorithm (4 marks)

End of questions

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